



Lesson 1: Valuation of Fixed Income Securities

Module : Valuation of Fixed Income Securities and Stocks

Introduction

In this lesson we will cover the following topics:

- Introduction to fixed income securities (FIS)
- Valuation of FIS through DCF methods
- Theories of term structure of interest rates
- Concept of yield to maturity
- Duration of an FIS and interest rate risk
- Summary and concluding remarks



Simple Valuation of Fixed Income Securities (FIS)

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If you own a fixed income security like a bond you are entitled to fixed set of payoffs called interest or coupons; and at maturity you get the face value or the principal

- Consider a simple bond that pays 8.5% interest. If you have invested \$100, you will get \$8.50 annually, if the coupons are annual, and at maturity you will also get the principal amount, i.e., total \$108.5. Also assume a 3% discount rate
- The PV of this bond can be easily computed as provided here
- $PV = ??$

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- Consider a simple bond that pays 8.5% interest. If you have invested \$100, you will get \$8.50 annually for the next four years, if the coupons are annual, and at maturity you will also get the principal amount, i.e., total \$108.5. Also assume a 3% discount rate
- The PV of this bond can be easily computed as provided here
- $$PV = \frac{8.50}{1.03} + \frac{8.50}{1.03^2} + \frac{8.50}{1.03^3} + \frac{108.50}{1.03^4} = \$120.44$$

Simple Valuation of Fixed Income Securities (FIS)

Given our knowledge of valuing annuities, we can separate the coupon payments and principal payment and compute the value of this bond in the manner as provided here

- $PV(\text{Bond}) = PV(\text{annuity of bond coupon payments}) + PV(\text{Principal payment})$
- $PV(\text{Bond}) = \frac{8.5}{0.03} * \left(1 - \frac{1}{1.03^4}\right) + \frac{100}{1.03^4} = 31.59 + 88.85 = \120.44
- The bond can be valued as a combination of (1) annuity, i.e., coupon payments and (2) a single final payment of principal

Simple Valuation of Fixed Income Securities (FIS)

Another very important concept for FIS is yield-to-maturity (YTM)

- In the previous example, if the bond under discussion, has a present value of \$120.44 then what is the current interest rate or yield of this bond to the buyer
- As the interest rates rise, bond prices fall and vice-versa
- This can be easily understood from the discounted cashflow valuation technique
- In this case, the answer is $ytm=3\%$, since we assumed the discount rate of 3% at the beginning

Simple Valuation of Fixed Income Securities (FIS)

This YTM can be easily computed as with expression shown here

- $120.44 = \frac{8.50}{1+ytm} + \frac{8.50}{(1+ytm)^2} + \frac{8.50}{(1+ytm)^3} + \frac{108.50}{(1+ytm)^4}$
- In this case, the answer is $ytm=3\%$, since we assumed the discount rate of 3% at the beginning
- If you buy this bond and hold it to maturity, you will earn a return of 3% per annum
- Why this bond has a ytm that is less than its coupon of 8.5%?
- You are paying \$120.44 for a bond with a face-value of \$100

Simple Valuation of Fixed Income Securities (FIS)

Consider the example of a simple bond with the following cash-flow profile

T=6m	T=12m	T=18m	T=24m	T=30m	T=36m
24.375	24.375	24.375	24.375	24.375	1024.375

- If the bond is currently trading at \$1107.95, then the current ytm of the bond can be simply computed from this equation provided here
- Coupons amounting to \$24.375 are paid semi-annually and at the end of the period, a principal payment of \$1000 is paid at the end of 3-years
- $PV = \frac{24.375}{1+\frac{ytm}{2}} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^2} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^3} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^4} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^5} + \frac{1024.375}{\left(1+\frac{ytm}{2}\right)^6}$; here $ytm/2=??\%$; $ytm=??\%$
- The effective annual yield (EAF) would be ?%

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- The effective annual yield (EAF) would be $(1.6003)^2-1=1.2042\%$



Bond Prices and Interest Rates

Bond Prices and Interest Rates

Bond prices change with interest rates

- In the previous example, where the semi-annual yield was 0.6003%, assume investors start demanding a semi-annual yield of 4%, that is annual percentage quoted rate of 8%
- The price of this bond will fall to reflect this change in yield, as per the computation shown here
- $$PV = \frac{24.375}{1+\frac{ytm}{2}} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^2} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^3} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^4} + \frac{24.375}{\left(1+\frac{ytm}{2}\right)^5} + \frac{1024.375}{\left(1+\frac{ytm}{2}\right)^6}$$

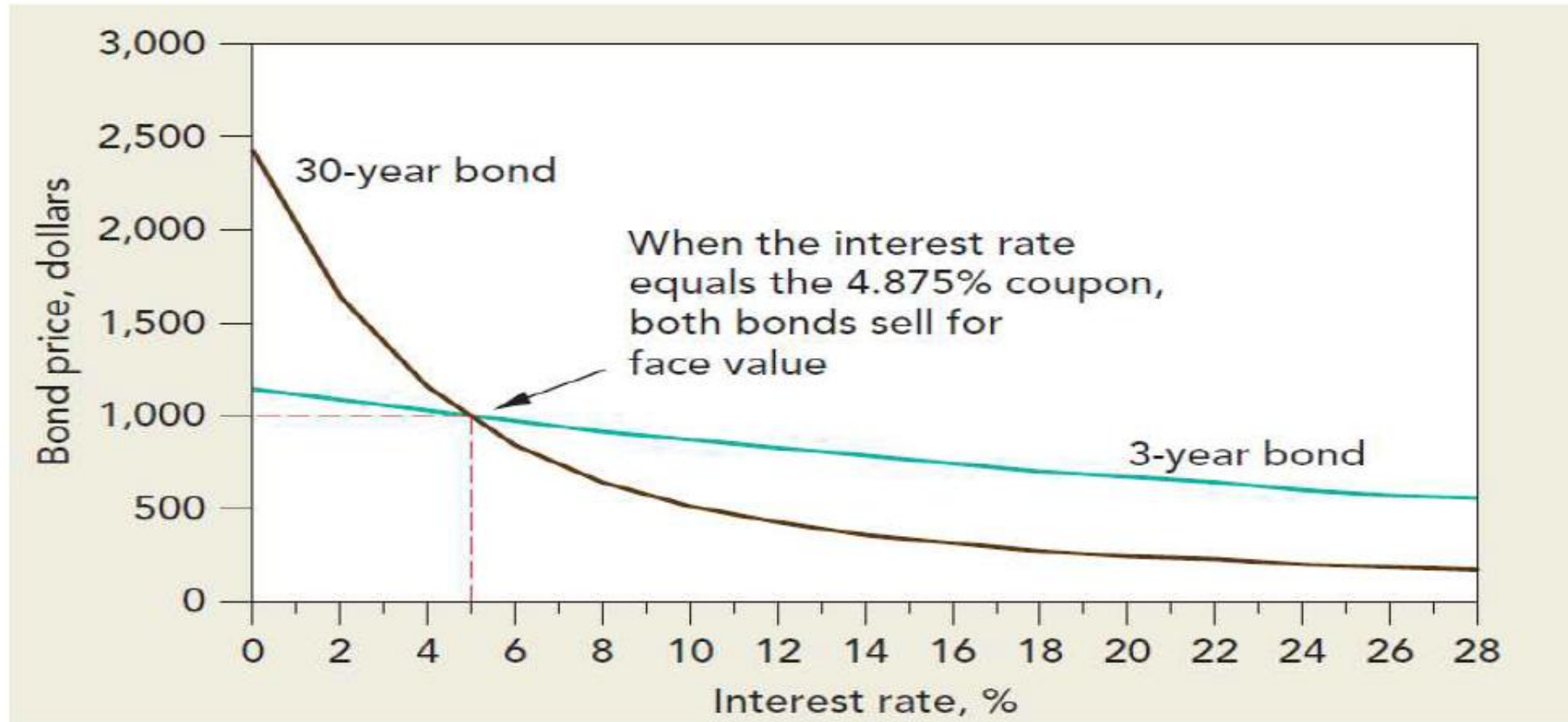
Bond Prices and Interest Rates

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- The price of this bond will fall to reflect this change in yield, as per the computation shown here
- $$PV = \frac{24.375}{1.04} + \frac{24.375}{(1.04)^2} + \frac{24.375}{(1.04)^3} + \frac{24.375}{(1.04)^4} + \frac{24.375}{(1.04)^5} + \frac{1024.375}{(1.04)^6} = \$918.09$$

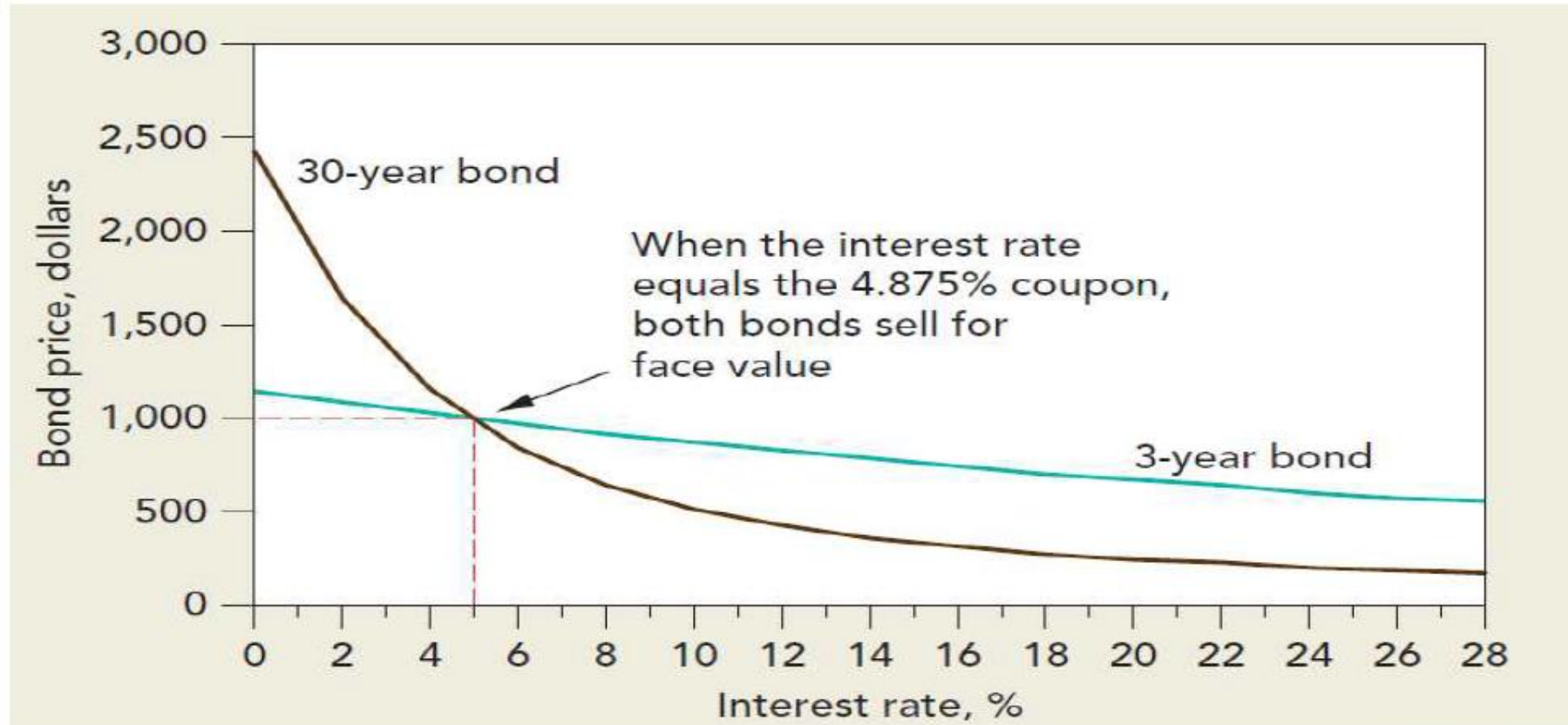
Bond Prices and Interest Rates

The cash flow discounting relation makes it quite obvious that with rising interest rates, prices would fall, and with falling interest rates, prices would rise



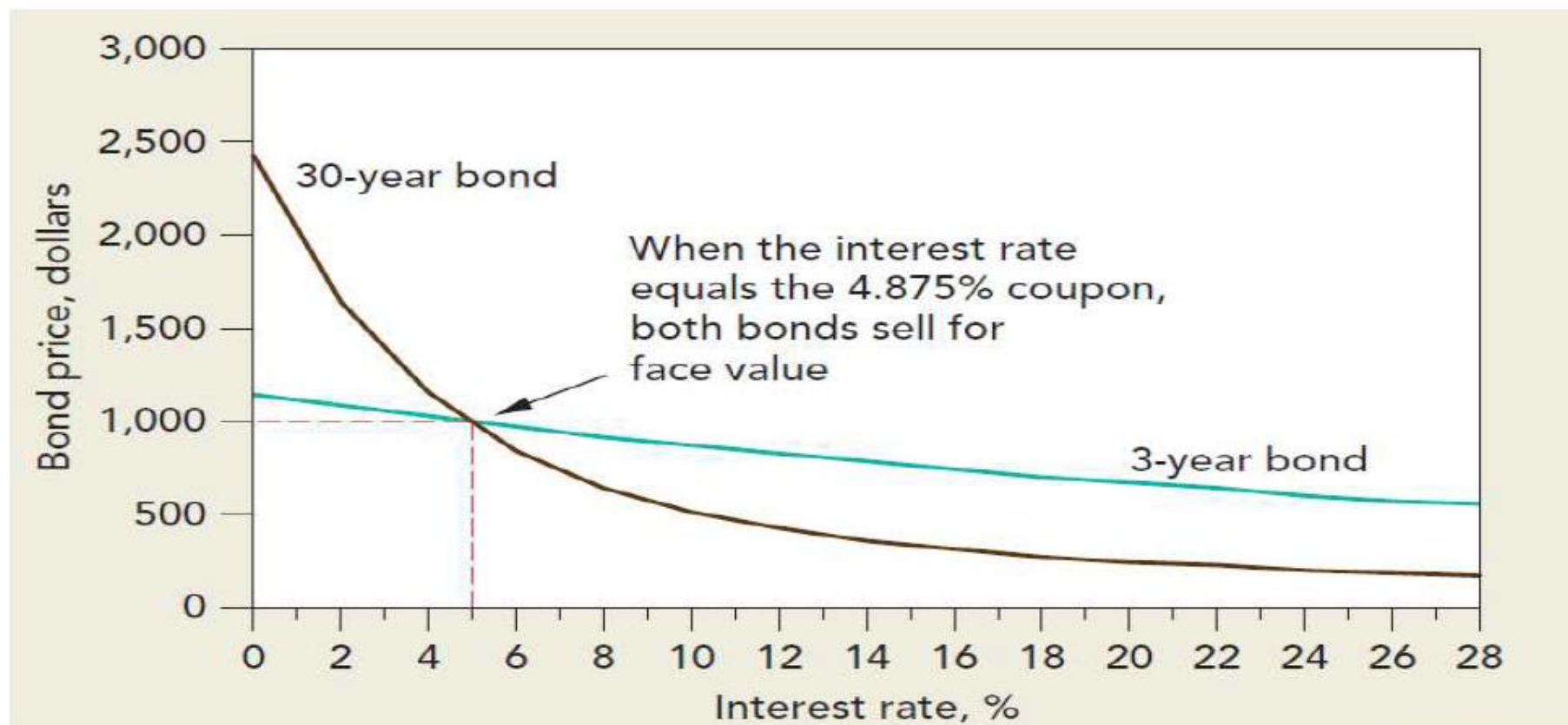
Bond Prices and Interest Rates

Let us examine the price-interest relationship for two bonds with the same coupon of 4.875%



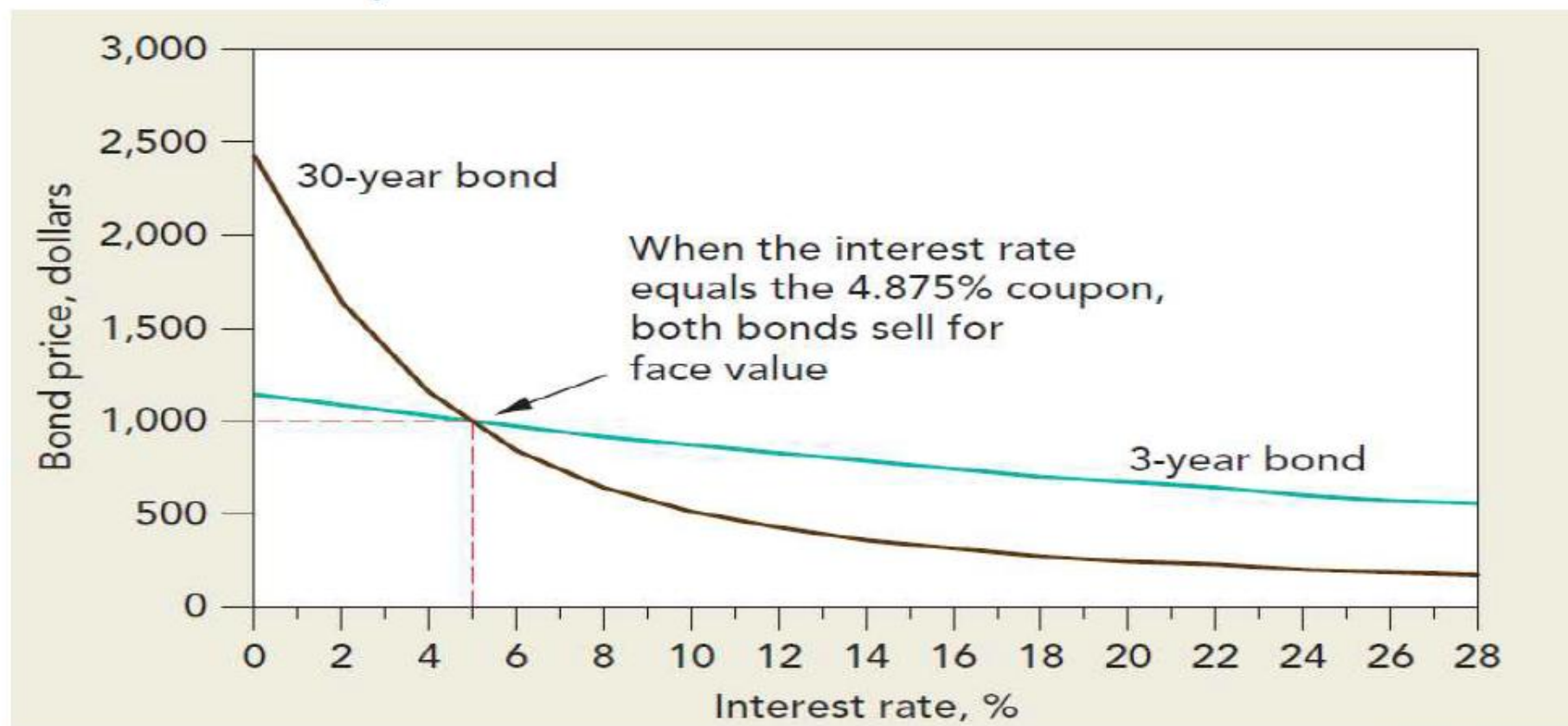
Bond Prices and Interest Rates

The impact of interest rate is only modest on near-term cash flows while it is more drastic on distant cash flows



Bond Prices and Interest Rates

This is precisely the reason why the price of a 30-year bond is much more sensitive to interest rate movements than a three-year bond





Duration of a Bond

Duration of a Bond

We saw that changes in interest rates have greater impact on the prices of long-term bonds than short-term bonds

- However, without including cash-flow profile, short-term and long-term is just a vague reference
- For example, compare a 30-year maturity bond with cash flows evenly spread over time with zero coupon STRIPS
- Separate Trading of Registered Interest and Principal Securities (STRIPS) are special instruments, created by stripping the cash flows from treasury instruments and government securities
- These are often called as zero-coupon bonds, and have the maturity same as duration

Duration of a Bond

It would be misleading to say that both of these instruments have the same maturity

	Price (%)	Cash payments %			
Bond	Feb. 2015	Aug. 2015	Feb. 2016...	... Aug. 2016	Feb. 2021
Strip for Feb. 2015	88.74	0	0 ...	... 0	100.00
Feb. 2015 (4% p.a.)	111.26	2.00	2.00 ...	... 2.00	102.00
Feb. 2015 (11.25% p.a.)	152.05	5.625	5.625 ...	... 5.625	105.625

- Consider three bonds, one strip and two coupon paying bonds with cash flow profile as provided here
- All of these bonds have a ytm of 2%
- Let us examine their time-pattern of cash flows
- The two coupon paying bonds offer a considerable proportion of their cash flows earlier than maturity.

Thus it is very easy to observe that the strip has the longest duration

Duration of a Bond

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Feb. 2015 (11.25% p.a.)	152.05	5.625	5.625 ...	... 5.625	105.625

- Bond with 11.25% coupon (i.e., 5.625% semi-annual coupon) offers a larger proportion of cash flows earlier than maturity, as compared to the bond with lower coupon of 4% (i.e., 2% semi-annual coupon)
- The bond with 11.25% coupon offers a larger proportion of cash flows earlier than maturity, as compared to the bond with a lower coupon of 4%
- If these two bonds are compared, the average maturity in cash flow terms should be higher for the 4% bond

Duration of a Bond

However, we need a more concrete measure of duration for which we have developed the intuition in the previous discussions

- The duration measure also indicates the sensitivity of a fixed income security to interest rate changes
- The simple measure of duration is computed as a weighted average of times, with weights being the present value of cash flows received at these times
- Consider a bond with a maturity of T years. The corresponding cash flows in each of these years being $C_1, C_2, \dots, C_T$ being received at the end of year 1, 2, 3, ..., T
- $$Duration = 1 * \frac{PV(C_1)}{PV} + 2 * \frac{PV(C_2)}{PV} + 3 * \frac{PV(C_3)}{PV} + \dots + T * \frac{PV(C_T)}{PV}$$

Duration of a Bond

Let us understand this through one example

- Consider a fixed income security with coupons of \$8.5 paid at the end of each year and a final principal payment in the final year, that is fourth year
- Also assume the appropriate interest rate of 3%

Year (t)	1	2	3	4	
Cash payment (C t)	8.5	8.5	8.5	108.5	PV
PV(Ct) at 3%	8.25	?	?	?	?
Fraction of total value [PV(Ct)/PV]	0.069	?	?	?	Total=Duration
Year x Fraction of total value [t x PV(Ct)/PV]	0.069	?	?	?	?

- $Modified\ Duration = \frac{Duration}{(1+yield)}$

Duration of a Bond

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Cash payment (C t)	8.5	8.5	8.5	108.5	PV
PV(Ct) at 3%	8.25	8.01	7.78	96.4	120.44
Fraction of total value [PV(Ct)/PV]	0.069	0.067	0.065	0.8	Total=Duration
Year x Fraction of total value [t x PV(Ct)/PV]	0.069	0.134	0.195	3.2	3.6 years

- $Modified\ Duration = \frac{Duration}{(1+yield)}$

Duration of a Bond

This modified duration measures the percentage change in price a one percentage change in yield (or interest rates)

- For our bond of duration 3.6 years. This measure works out to $3.6/1.03 = 3.49\%$
- Now consider a scenario where interest rates rise by 0.5% and fall by the same amount

Year (t)	1	2	3	4	PV	Change (%)
Cash payment (C_t)	8.5	8.5	8.5	108.5		
PV(C_t) at 3%	8.25	8.01	7.78	8.01	120.44	
PV(C_t) at 3.5%	8.21	?	?	?	?	A
PV(C_t) at 2.5%	8.29	?	?	?	?	B

- The total magnitude of change works out to $A\% + B\% = 3.49\%$
- This is the same amount as our modified duration measure

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PV(Ct) at 3.5%	8.21	7.93	7.67	94.55	118.37	-1.72%
PV(Ct) at 2.5%	8.29	8.09	7.89	98.30	122.57	1.77%

- The total magnitude of change works out to $1.72\% + 1.77\% = 3.49\%$
- This is the same amount as our modified duration measure

Duration of a Bond

Duration is an important measure of sensitivity of an instrument to changes in interest rates

- As the duration of an instrument increases, its sensitivity to interest rate changes also increases
- The interest rate risk of fixed income securities is computed with the help of this duration measure
- An important application of this duration measure is in managing the interest rate risk of portfolios

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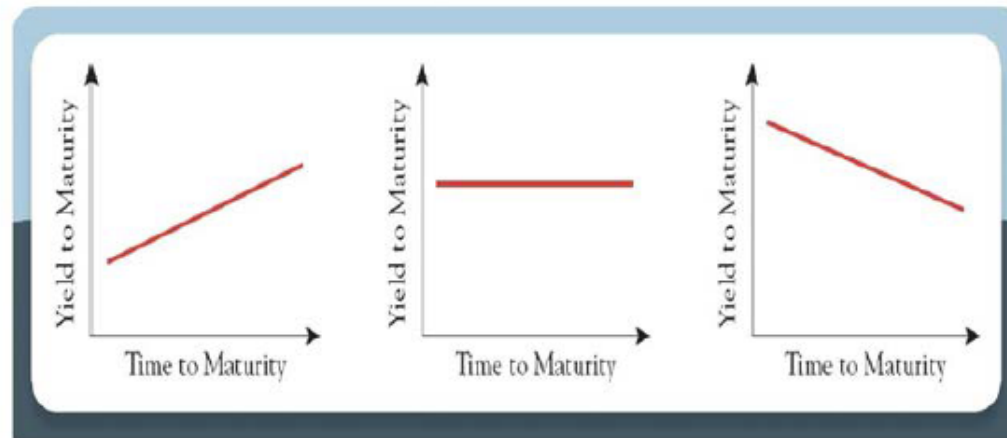
- Portfolio managers often rely on this duration measure for the fixed income securities

Term Structure of Interest Rates

Term Structure of Interest Rates

Till now we have used the same interest rates to discount the cash-flows irrespective of the tenors

- This assumes single yield to maturity across different tenors
- Interest rates vary over different tenors, and short-term interest rates are different from long-term interest rates



Interest rates are expected to rise.

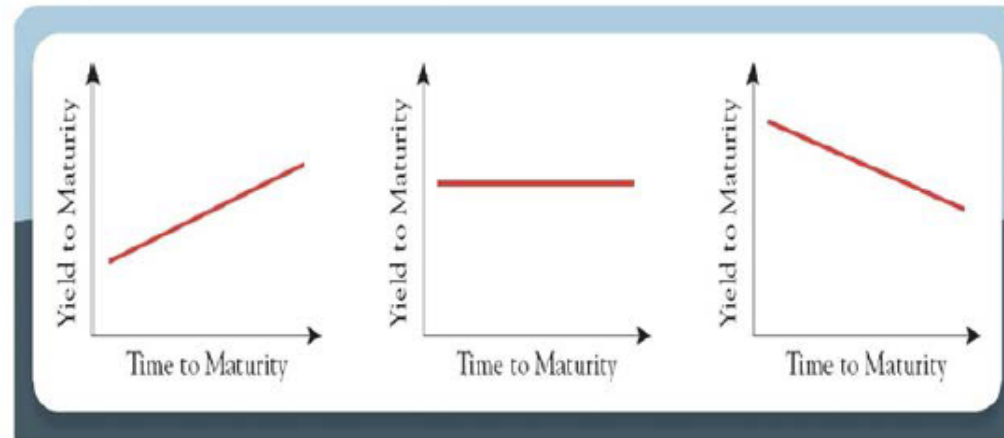
Interest rates are expected to remain unchanged.

Interest rates are expected to fall.

Term Structure of Interest Rates

This variation in interest rates over short-term and long-term and across periods, is often referred to as term structure of interest rates

- This term structure is most often rising, but sometimes can be falling as well
- The figure here shows all the possible term structures, i.e., rising, falling, and constant



Interest rates are expected to rise.

Interest rates are expected to remain unchanged.

Interest rates are expected to fall.

Term Structure of Interest Rates

Consider a term structure of interest rates $r_1, r_2, r_3, \dots, r_t$ for time periods 1, 2, 3, ..., t

- A simple cash-inflow of \$1 in the first year will have a value of $PV = \frac{1}{1+r_1}$. Here r_1 , would be called the one-year spot rate
- Similarly, a loan that pays \$1 at the end of two years, will have a present value of $PV = \frac{1}{(1+r_2)^2}$
- For simple illustration purposes assume that $r_1 = 3\%$ and $r_2 = 4\%$. A security that offers only these two cash flows will have a present value of $PV = \frac{1}{1.03} + \frac{1}{(1.04)^2} = 1.895$
- $PV = \frac{1}{1+ytm} + \frac{1}{(1+ytm)^2} = 1.895$
- Solving for this equation, we get ytm= ?%

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- $PV = \frac{1}{1+ytm} + \frac{1}{(1+ytm)^2} = 1.895$
- Solving for this equation, we get ytm= 3.67%

Term Structure of Interest Rates

In a well-functioning liquid and efficient markets, all safe (that is risk-free cash flows) must be discounted at the same risk-free spot: Law of one price

		1	2	3	4	Bond Price (PV)	Ytm
	Spot rates	3.50%	4%	4.20%	4.40%		
	Discount factor	0.97	0.92	0.88	0.84		
A	8% coupon-2year	80	1080				
	PV	77.29	998.52	-	-	1,075.82	?%
B	11%-coupon-3year	110	110	1110			
	PV	?	?	?	-	?	?%
C	6% coupon-4year	60	60	60	1060		
	PV	?	?	?	?	?	?%
D	STRIP				1000		
					?	?	?%

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	PV	106.28	101.70	981.11	-	1,189.10	4.16%
C	6% coupon-4year	60	60	60	1060		
	PV	57.97	55.47	53.03	892.29	1,058.76	4.37%
D	STRIP				1000		
					841.78	841.78	4.40%

Term Structure of Interest Rates

Consider four risk-free government bonds, with annual coupon payments as shown here

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Term Structure of Interest Rates

Bond A is the shortest duration bond among all, while the STRIP D is of the longest duration

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Term Structure of Interest Rates

The law of one price says that each of these risk-free dollars should carry the same value

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Term Structure of Interest Rates

For example, \$1 received in year 4 has the same value whether from bond A, B, C, or D

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Term Structure of Interest Rates

This also means that one can use the same discount factor for each of the years (1, 2, 3, 4) for all the bonds

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Term Structure of Interest Rates

Once the PV of these bonds is computed, ytm can also be easily computed

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Term Structure of Interest Rates

The procedure to estimate the spot rates for different maturities, and therefore, the term structure of interest is not too difficult

- All you need is a bond with single cash flow (i.e., STRIPS) for different maturities
- Using the price of these STRIPS, once can estimate the spot rates prevailing in the market for that maturity
- For example, consider A 10-year strip with face-value of \$1000 at then end of maturity is selling at \$714.18
- This also means that for each \$1 to be received after 10-years, investors are willing to pay \$0.714
- $F_0 = \frac{1}{(1+r_{10})^{10}} = 0.71418$; solving for this, $r_{10} = 3.42\%$

Term Structure of Interest Rates

Expectations theory of term structure: Term-structure of interest rates reflect the expectation of interest rates in future

- A rising term structure indicates that market participants expect the interest rates to rise in future
- Assume that the spot rate for year 1, r_1 is 5% and spot rate for year 2, r_2 is 7%
- If you invest \$100 for one year, you get \$5 for interest. If you invest it for two years, you get $100 * 1.07^2$, that is, \$114.49 after two years
- The extra return that you earn in second year can be computed as noted here: $\frac{1.07^2}{1.05} - 1 = 9.0\%$
- This means that if you invest for two years, you will get 5% in year 1 and 9% in year 2

Term Structure of Interest Rates

If you expect that bond prices in the year 2 will yield more, then you would prefer to invest at 1-year spot and then invest in second year at prevailing rate

- If many people would think like that, they will sell the second year bond and buy the first year
- Ultimately, prices of these bonds have to adjust so that people are equally satisfied whether they invest in 1 year spot or 2 year spot
- In equilibrium, long term spot rates are a combination of short-term spot and a series of forward rates
- Forward rates are future rates booked (contracted) today. For example, rate of interest for period $T=1$ to $T=2$ booked at $T=0$; or interest rate for $T=2$ to $T=4$ booked at $T=0$

Term Structure of Interest Rates

Liquidity preference theory suggests that investors prefer to invest in short-term as they fear the additional volatility, risk, and uncertainty associated with the long-term instruments

- Investors require additional risk-premia to hold long-term instruments as against short-term instruments
- Both the expectations theory and liquidity preference theory are employed to explain the dynamics of term-structure of interest rates
- To summarize, interest rates might differ for different maturities, they can be rising or falling
- This variation in interest rates across different maturities leads to a very important concept called the term structure of interest rates

Summary and Concluding Remarks

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- Fixed income securities like bonds are simply long-term loans
- These instruments include regular interest (or coupon) payments and at the maturity you get back the face-value (or principal)
- These instruments can be easily valued through discounting cash flow valuation method
- Also, it is appropriate to discount each of these cash flows with its on spot rate corresponding to the duration of the cash flows
- The spot rate is observed on the term structure of interest rates. The term structure of interest rates is computed using the STRIPs

Summary and Concluding Remarks

- Once the present value of a bond is computed, using bond cash flows, one can also calculate the ytm of the bond
- Duration reflects the average time associated with cash-flows of a fixed income security
- The expectations theory of interest rates suggests that rising interest rates reflect the future expectations of investors
- The theory of liquidity preference suggests that investors prefer to hold short-term instruments as compared to long-term instruments

Thanks!